KEY SHEET

PHYSICS

1	**BCD**	2	**AC**	3	**CD**	4	**ACD**	5	**AC**	6	**ABC**
7	**12**	8	**100**	9	**0.25**	10	**121.14-121.15**	11	**201.60**	12	**10.08**
13	**B**	14	**B**	15	**A**	16	**A**	17	**2**	18	**1**
19	**8**										

CHEMISTRY

20	**AC**	21	**ABCD**	22	**ACD**	23	**ABD**	24	**ABC**	25	**ABC**
26	**2**	27	**4**	28	**172.41**	29	**1.15**	30	**101**	31	**150.55**
32	**D**	33	**D**	34	**C**	35	**D**	36	**3**	37	**3**
38	**7**										

MATHEMATICS

39	**BC**	40	**ABCD**	41	**AD**	42	**BC**	43	**BD**	44	**ABD**
45	**1**	46	**2**	47	**12**	48	**21**	49	**4**	50	**3**
51	**B**	52	**D**	53	**D**	54	**A**	55	**4**	56	**2**
57	**90**										

SOLUTIONS
PHYSICS

1) Conceptual
2) Conceptual
3) Conceptual
4) Conceptual
5) Conceptual
6) Conceptual
7) Conceptual
8) Conceptual
9) Conceptual
10) Conceptual
11) Conceptual

Ans (10.08)

$$F = QE = Q\left(\frac{\rho R}{6\epsilon_0}\right)$$

$$= \frac{QR}{6\epsilon_0}\left(\frac{Q}{\frac{4\pi}{3}\left(R^3 - \frac{R^3}{8}\right)}\right) = \frac{Q^2}{7\pi\epsilon_0 R^2}$$

12)
13) Conceptual
14) Conceptual
15) Conceptual
16) Conceptual
17) Conceptual
18) Conceptual
19)

Ans (8)

For the first question, apply the Boltzmann equatio

and

$$P = \sigma A T_s^4$$

where A is the surface area of the planet, and T_s th temperature at the center. Then

$$T_s = \left(\frac{P}{4\pi\sigma R^2}\right)^{\frac{1}{4}}$$

For the second question, it is reasonable to assur that the temperature depends on the distance form th center only. Then the definition of k gives for spherical shell of thickness dr

$$k = \frac{\Delta Q}{\Delta t}\frac{1}{4\pi r^2}\frac{dr}{dT}.$$

The heat through the shell depends on the power radiated from within the shell. Since the planet is uniform, this depends on the volume according to

$$\frac{\Delta Q}{\Delta t} = P\frac{\frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} = P\frac{r^3}{R^3}$$

so that rearrangement yields

$$dT = \frac{P}{4\pi k R^3} r\, dr$$

Integrating between the center and the surface,

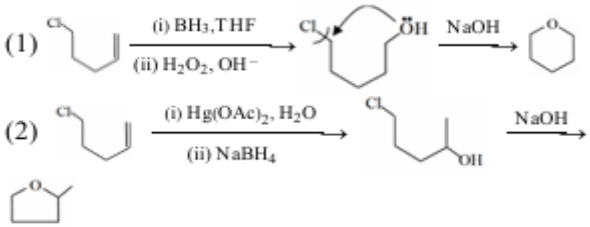
$$\Delta T = \frac{P}{8\pi k R},$$

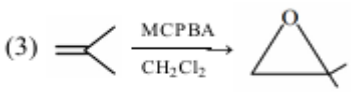
which could be used to find the temperature of the interior.

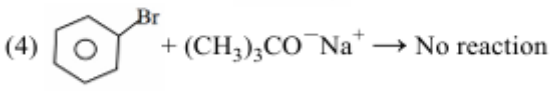
CHEMISTRY

20.

Ans (A,C)

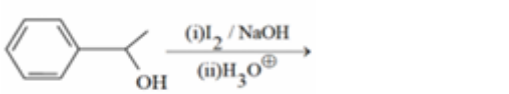


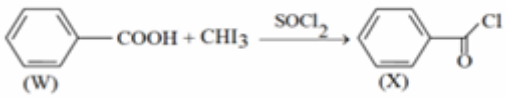


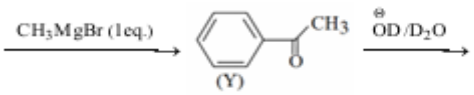


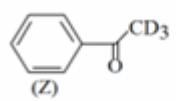
21.

Ans (A,B,C,D)









22.

Ans (A,C,D)

$$\underbrace{K_2Zn_3[\overset{+2}{Fe}(CN)_6]_2}_{\text{diamagnetic}} \xrightarrow{NaOH} K^+ + Zn(OH)_4^{2-} + [Fe(CN)_6]^{4-}$$

$\hookrightarrow$ Synergic bonding b/w Fe^{+2} and $\overline{C}N$.

23.

Ans (A,B,D)

$X = XeOF_4$, $Y = XeO_2F_2$, $Z = XeO_3$

24.

Ans (A,B,C)

M = Cu

$$Cu^{2+} \xrightarrow{K_4[Fe(CN)_6]} \underset{\text{(Brown)}}{Cu_2[Fe(CN)_6]} \downarrow$$

$$E^{\circ}_{Cu^{+2}/Cu} = \text{'+'Ve}$$

25.

Ans (A,B,C)

(A) pH = 5 + log 1 = 5 (Buffer solution)

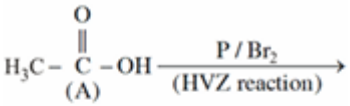
(B) salt hydrolysis

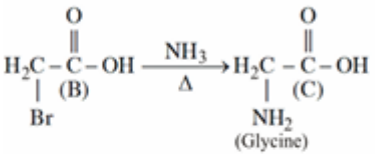
(C) $[H^+] = \sqrt{CKa} = 10^{-5}$; pH = 5

(D) $[OH^-] = 2 \times 5 \times 10^{-6} = 10^{-5}$, pOH = 5, pH = 9

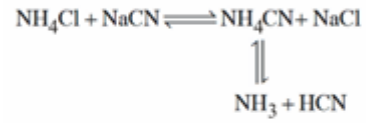
26.

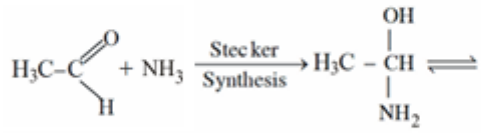
Acid (A) is ethanoic acid as it is an important constitutent of vineagar.

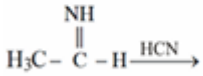


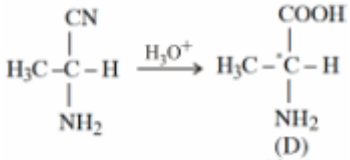


Also,









Now, since (C) has no chiral carbon. So, x = 0

And (D) has one asymmetric carbon, so y = 2

27.

Ans (4.00)

Total number (z) of dipeptides which can be made by (C) and (D) are

gly - gly, ala - ala, gly - ala, ala - gly

Hence, z = 4

28.

Ans (172.41)

$$Cv_{mix} = \left(\frac{n_1Cv_1 + n_2Cv_2}{n_1 + n_2}\right)$$

$$= \frac{\left(0.5 \times \frac{5R}{2}\right) + \left(0.5 \times \frac{5R}{2}\right)}{1}$$

$$Cv_{mix} = \frac{5R}{2}$$

$$Cp_{mix} = R + Cv_{mix}$$

$$= R + \frac{5R}{2}$$

$$= \frac{7R}{2}$$

For reversible adiabatic process

$$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1}$$

$$300(1)^{\gamma-1} = t_2(4)^{\frac{7}{5}}$$

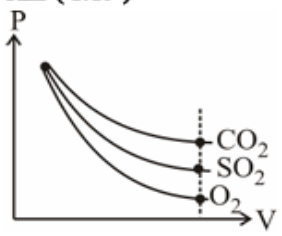
$$300 \times 1 = T_2(4)^{\frac{2}{5}}$$

$$300 = T_2 \times 1.74$$

$$T_2 = (172.41\ K)$$

29.

Ans (1.15)



$X = O_2 \quad \gamma = \frac{9}{7} = 1.28$

$Y = CO_2 \quad \gamma = \frac{15}{13} = 1.15$

$Z = SO_2 \quad \gamma = \frac{7}{6} = 1.16$

30.

Ans (101.00)

$G = NaHCO_3$, $D = NH_3$

31.

32.

Ans (D)

Rate of diffusion of methane

$$= \frac{1}{16 \times 20} \text{mole sec}^{-1}$$

Rate of diffusion of hydrocarbon

$$= \frac{\sqrt{20}}{M \times 40} \text{moles sec}^{-1}$$

Now, $\frac{r_{CH_4}}{r_{hyd}} = \sqrt{\frac{M_{hyd}}{16}}$

$$\frac{\frac{1}{16\times20}}{\frac{\sqrt{20}}{M\times40}} = \frac{\sqrt{M}}{4} \Rightarrow \frac{40 \times M}{16 \times 20 \times \sqrt{20}} = \frac{\sqrt{M}}{4}$$

$$\Rightarrow \sqrt{M} = 2\sqrt{20}$$

$$\Rightarrow M = 80\ g\ mol^{-1}$$

0.125 m mol of (A) consumes 0.375 m mol H_2. So,

(A) has three π-bonds

Therefore : molecular formula of A is C_6H_8

An open chain isomer of A can be

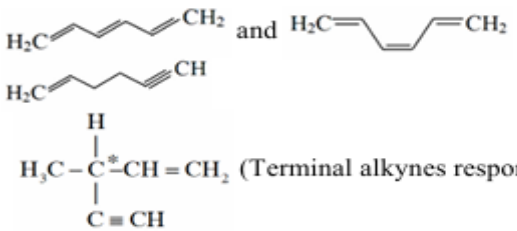
$CH_2 = CH - CH = CH - CH = CH_2$

A six membered stable monocyclic isomer of (A)

33.

Ans (D)

$$\Rightarrow = M = 80\ g\ mol^{-1}$$



(Terminal alkynes responsed to tollen's test)

34. & 35.

Ans (D)

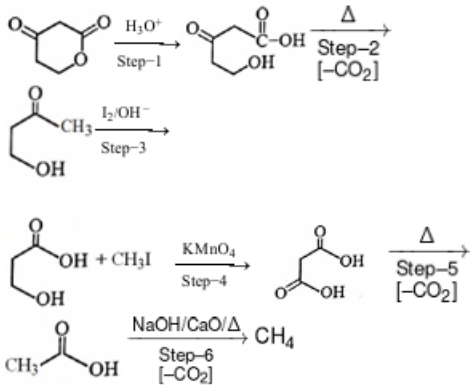
$$\underset{1\text{mole}}{NaN_3(s)} \xrightarrow[T<300^\circ C]{\Delta} \underset{(\text{solid})}{Na} + \underset{(\text{gas})}{\frac{3}{2}N_2}$$

$$N_{2(g)} + 3H_2 \rightarrow 2NH_3(g)$$

$$NH_3 + Ca(OCl)Cl \longrightarrow N_2 + CaCl_2 + H_2O$$

36.

Ans (3)



37.

Ans (3)

$\Delta T_b = i.K_b.m$

$0.08 = i \times 0.5 \times \frac{2}{20}$

$\frac{0.08}{0.5} \times \frac{20}{2} = i$

$i = 1.6$

$\Lambda_m = \Lambda_m^\circ.C.\alpha.$

$= 500\frac{Scm^2}{mole} \times \frac{2}{20}\frac{mole}{L} \times 0.6$

$\Lambda_m = 30\frac{Scm^2}{1000cm^3} = \frac{3}{100}Scm^{-1}$

$\Lambda_m = 3\ Sm^{-1}$

38.

Ans (7)

$BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 \downarrow +2NaCl$

0.2 M 0.1 M ppt

0.04 mole 0.02 mole 0.08 mole

$\pi_{Na_2SO_4} = i_1C_1RT \qquad \pi_{NaCl} = i_2C_2RT$

$\pi_{Na_2SO_4} + \pi_{NaCl} = (i_1C_1 + i_2C_2).RT$

$= 3 \times \left(\frac{0.02}{0.8}\right) + 2\left(\frac{0.08}{0.8}\right).R \times T$

$= (0.075 + 0.2)RT$

$= 0.275 \times 0.0821\ atm \times 300$

$= 6.99$

$= 7\ atm$

MATHEMATICS

39.

$xf(y) < yf(x)\ \forall\ 0 < x < y < 1$

$\Rightarrow \quad \dfrac{f(y)}{y} < \dfrac{f(x)}{x}\ \forall x < y$

$\Rightarrow \quad g(x) = \dfrac{f(x)}{x}$ is decreasing

$\Rightarrow \quad g'(x) < 0 \Rightarrow xf'(x) < f(x)$

$\Rightarrow \quad f'(x) < \dfrac{f(x)}{x} \forall\ x, 0 < x < 1$

Hence, $f'(x) < g(x) \forall\ x, 0 < x < 1$

Hence, f '(x) is less than minimum value of g(x)

which is g (1)

$\therefore\ f'(x) < f(1)$

$x\ f'(x) < f(x) \Rightarrow \displaystyle\int_0^1 x\, f'(x)\, dx < \int_0^1 f(x)\, dx$

$\Rightarrow x\ f(x)\ |_0^1 - \int_0^1 f(x)\ dx < \int_0^1 f(x)\ dx$

$\Rightarrow \quad f(1) < 2\displaystyle\int_0^1 f(x)\ dx$

40.

$|z_1\ z_2| = \left|\dfrac{c}{a}\right| = 1, \quad |z_1 + z_2| = \left|\dfrac{-b}{a}\right| = 1$

$\Rightarrow \quad (z_1 + z_2) \times (\bar{z}_1 + \bar{z}_2) = 1$

$\Rightarrow (z_1 + z_2)\left(\dfrac{1}{z_1} + \dfrac{1}{z_2}\right) = 1$

$\Rightarrow \quad (z_1 + z_2)^2 = z_1\ z_2$

$\Rightarrow \quad \left(\dfrac{-b}{a}\right)^2 = \dfrac{c}{a} \Rightarrow b^2 = ac$

$|z_1 + z_2| = |z_1||1 + e^{i\theta}| = 2\cos\dfrac{\theta}{2} = 1 \Rightarrow \theta = \dfrac{2\pi}{3}$

$PQ = |z_2 - z_1| = |z_1|\,|e^{i\theta} - 1| = 2\sin\dfrac{\theta}{2}$

$$PQ = |z_2 - z_1| = |z_1||e^{i\theta} - 1| = 2\sin\frac{\theta}{2}$$

$$= 2\sin\frac{\pi}{3} = \sqrt{3}$$

41.

(a) $T_n^3 + T_n = T_{n+1} - T_n$

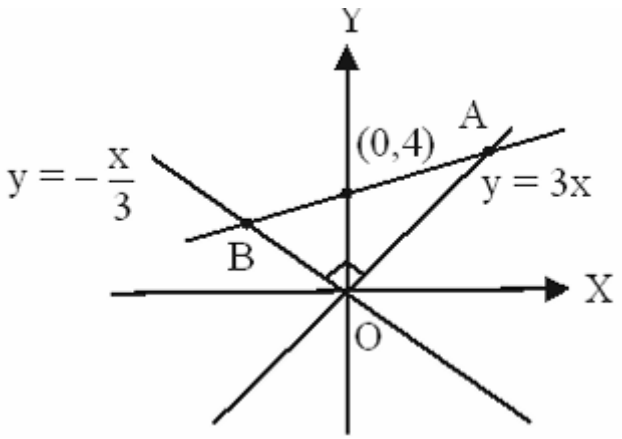
$$\sum_{n=1}^{100}(T_n^3 + T_n + 1) = T_{101} - T_1 + 100 = T_{101} + 99$$

(c) $T_n^2 + 2 = \frac{T_{n+1}}{T_n}$

$$\prod_{n=1}^{100}(T_n^2 + 2) = \prod_{n=1}^{100}\frac{T_{n+1}}{T_n} = \frac{T_{101}}{T_1} = T_{101}$$

(d) $\prod_{n=1}^{100}\left(\frac{T_{n+1}}{T_n} - T_n^2\right) = 2^{100}$

42.



$(3x-y)(x+3y) = 0$

$y = 3x, y = \frac{-x}{3}$

(a) $m_1 + m_2 = 3 - \frac{1}{3} = \frac{8}{3}$

(b) $am_1 + m_2 = 3\left(-\frac{1}{3}\right) + 3 = 2$

(c) $y = 2x + 4$

$A \equiv (4, 12), B = \left(\frac{-12}{7}, \frac{4}{7}\right)$

$$\text{Area}(\Delta AOB) = \frac{1}{2} \times \sqrt{16+144} \times \sqrt{\frac{144+16}{49}} = \frac{160}{2\times 7} = \frac{80}{7}$$

43.

$(A^2 - 2I)B = 0 \Rightarrow A^2 = 2I$ and

$B = \text{adj } A = |A|.\frac{A}{2} = -\sqrt{2}A \{\because |A| = -2\sqrt{2}\}$

$AB = A(\text{adj } A) = |A|.\ I$

$\text{tr.}(AB) = 3|A| = -6\sqrt{2}$

and $\det.(A - \sqrt{2}B) = \det.(A + 2A) = \det.(3A)$

$= 27(-2\sqrt{2}) = -54\sqrt{2}$

44.

$x^3 - 3x + \lambda = 0$ let root $= \alpha, \beta, \gamma$

$\therefore\ \alpha + \beta + \gamma = 0,\ \alpha\beta + \beta\gamma + \gamma\alpha = -3,\ \alpha\beta\gamma = -\lambda$

$\therefore\ \alpha^2 + \beta^2 + \gamma^2 = 6,\ (\alpha, \beta, \gamma \in I)$

$\therefore$ Possibilities are

$$\left.\begin{matrix}\alpha = 2, \beta = -1, \gamma = -1) \\ \alpha = -2, \beta = 1, \gamma = 1\end{matrix}\right\} \Rightarrow \lambda = 2 \text{ or } -2 \Rightarrow (\lambda_1)^2 + (\lambda_2)^2 = 8$$

$\therefore\ h(x) = x^3 - 3x + 4$

Required area

$$= \int_2^5 h(x)\,dx = \int_2^5 (x^3 - 3x + 4)\,dx = \frac{531}{4}$$

And $\int_1^3 (x^3 - 3x)\,dx + \int_2^{22} g(x)\,dx = \int_1^3 (x^3 - 3x + 4)\,dx + \int_2^{22} g(x)\,dx - 8$

$= 66 - 2 - 8 = 56$

45.

a, c, b are in H.P, $\Rightarrow \frac{2}{c} = \frac{1}{a} + \frac{1}{b}$

Family of lines

$\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0 \Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right) = 0$

$\Rightarrow \frac{1}{a}\left(x + \frac{1}{2}\right) + \frac{1}{b}\left(y + \frac{1}{2}\right) = 0$

$\Rightarrow$ centre of ellipse $\equiv \left(\frac{-1}{2}, \frac{-1}{2}\right)$

$AC : x^2 + y^2 + 2px + 2qy - 8p + \frac{1}{2} = 0$

$\Rightarrow p = \frac{1}{2} : q = \frac{1}{2}$; radius of AC $= 2$

$\therefore\ \ell$ (major axis) $= 2a = 4$

$\Rightarrow$ We know, harmonic mean of segment of focal chord is equal to semi latus rectum.

semi latus rectum $= \frac{2.(1 \times 2)}{(1 + 2)} = \frac{4}{3}$

46.

a,c, b are in H.P, $\Rightarrow \frac{2}{c} = \frac{1}{a} + \frac{1}{b}$

Family of lines :

$\frac{x}{a} + \frac{y}{b} + \frac{1}{c} = 0 \Rightarrow \frac{x}{a} + \frac{y}{b} + \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right) = 0$

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AC : $x^2 + y^2 + px + 2qy - 8p + \frac{1}{2} = 0$

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$\therefore$ *l* (major axis) = 2a = 4

$\Rightarrow$ We know, harmonic mean of segment of focal chord is equal to semi latus rectum.

Semi latus rectum

$= \frac{2.(1 \times 2)}{(1+2)} = \frac{4}{3}$

$\therefore$ latus rectum, L = $\frac{8}{3} = 2.67$

[L] = [2.67] = 2

47.

Put $x = \cos\theta$

$1 - x^2 = \sin^2\theta. \Rightarrow \sqrt{x^2 - 1} = i\sin\theta$

$\left(x + \sqrt{x^2 - 1}\right)^{10} + \left(x - \sqrt{x^2 - 1}\right)^{10}$

$= (\cos\theta + i\sin\theta)^{10} + (\cos\theta - i\sin\theta)^{10}$

$= 2\cos 10\theta = 2f_{10}(x) = \lambda f_k(x)$

$\therefore \lambda = 2; k = 10 \Rightarrow \lambda + k = 12.$

48.

$f_6(\cos x) = \cos 6x$

we know,

$(\cos x + i \sin x)^6 = \cos 6x + i \sin 6x$

Equating real part both sides :

$\cos\ 6x = 6_{C_0}\cos^6 x - 6_{C_2}\cos^4 x \sin^2 x + 6_{C_4}\cos^2 x \sin^4 x - 6_{C_6}\sin^6 x$

$f_6(\cos x) = \cos 6x = \cos^6 x - 15\cos^4 x(1 - \cos^2 x) + 15$

$\cos^2 x(1 - \cos^2 x)^2 - (1 - \cos^2 x)^3$

$\therefore f_6(x) = x^6 - 15x^4(1 - x^2) + 15x^2(1 - x^2)^2 - (1 - x^2)^3$

$\therefore p = 32\ ; q = -48; r = 18, s = -1.$

$\therefore 7(p + q + r - s) = 7(32 - 48 + 18 + 1) = 7 \times 3 = 21$

49.

Equation of family of planes containing the line of intersection of P_1 and P_2 is given by $P_1 + \lambda P_2 = 0$

$\Rightarrow (1 + \lambda)x + (1 - 2\lambda)y + (2 + \lambda)z - 3 - 4\lambda = 0$

.....(1)

Projection of length of line segment AB is greatest, then AB must be parallel to plane (1)

$\therefore$ drs of AB = 2, – 1,3.

$2(1 + \lambda) - 1(1 - 2\lambda) + 3(2 + \lambda) = 0 \Rightarrow \lambda = -1$

$\therefore P_3 : 3y + z + 1 = 0. \Rightarrow a = 0, b = 3, c = 1$

50.

Projection of length of line segment is least if AB is perpendicular to plane (1).

$$\frac{1+\lambda}{2} = \frac{1-2\lambda}{-1} = \frac{2+\lambda}{3} \Rightarrow \lambda = 1$$

$\therefore$ Equation of plane P_4 is

$2x - y + 3z = 7. \Rightarrow a = 2, b = -1, c = 3$

51.

a, b and c are in A.P.

$\therefore\ \sqrt{ac} < b$

$$f(x) = \lim_{n\to\infty} \frac{-(e^x)^n + x^2 + f}{2e^x(e^x)^n + x + d} = \begin{cases} \frac{x^2+f}{x+d} & , \quad x < 0 \\ \frac{-1+f}{2+d} & , \quad x = 0 \\ \frac{-1}{2e^x} & , \quad x > 0 \end{cases}$$

For f(x) to be continuous

$$\frac{f}{d} = \frac{-1+f}{2+d} = \frac{-1}{2}$$

$\therefore\ 2f + d = 0 \ \Rightarrow\ 2f + d + 1 = 1$

52.

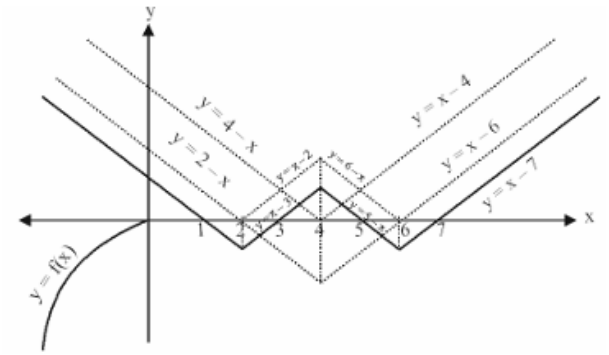
a, b and c are in G.P.

$$f(x) = \lim_{n\to\infty} \frac{x^2 + f}{2e^x(e^x)^n + x + d} = \begin{cases} \frac{x^2+f}{x+d} & , \quad x < 0 \\ \frac{f}{2+d} & , \quad x = 0 \\ 0 & , \quad x > 0 \end{cases}$$

For f(x) to be continuous

$$\frac{f}{d} = \frac{f}{2+d} = 0 \qquad \Rightarrow\ f = 0$$

$f(x) = ||x - 4| - 2| - 1$



No of solution = 4

53.

$$P\left(\frac{A}{B_E}\right) = \frac{P(A \cap B_E)}{P(B_E)}, \text{ where A = sum is 4.}$$

And $P(B_E) = P(B_2) + P(B_4) + \ldots\ldots\ldots = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \ldots\ldots\ldots\infty$

$$= \frac{\frac{1}{4}}{1 - \frac{1}{4}} = \frac{1}{4} \cdot \frac{4}{3} = \frac{1}{3}$$

$$P(A \cap B_E) = P(A \cap B_2) + P(A \cap B_4)$$

$$= P(B_2)P\left(\frac{A}{B_2}\right) + P(B_4)P\left(\frac{A}{B_4}\right)$$

$$= \frac{1}{4}\left(\frac{3}{36}\right) + \frac{1}{16}\left(\frac{1}{6^4}\right) = \frac{(4.3.6^2 + 1)}{16.6^4} = \frac{433}{16.6^4}$$

Hence $P\left(\frac{A}{B_E}\right) = \frac{433}{16.6^4}.3 = \frac{433}{32.216} \cong \frac{1}{16}$

54.

$n(S) = 6 \times 6 \times 6 = 6^3 = 216$

Now greatest number is 4, so atleast one of the dice shows up 4.

$\therefore\ n(A) = 4^3 - 3^3 = 37$

Hence $P(A) = \frac{37}{216}$

55.

$$\frac{1}{5}\left[\frac{1}{3} + \frac{1}{6} + \frac{1}{10} + \frac{1}{15} + \ldots\ldots\infty\right]$$

$$\frac{2}{5}\left[\frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \ldots\ldots\infty\right]$$

$$\frac{2}{5}\left[\frac{(3-2)}{6} + \frac{(4-3)}{12} + \frac{(5-4)}{20} + \ldots\ldots\infty\right]$$

$$\frac{2}{5} \times \frac{1}{2} = \frac{k}{20} \Rightarrow k = 4$$

56.

$$\frac{a}{\cos(200^\circ)} - \frac{c\sin(200^\circ)}{\cos(200^\circ)} = d \quad \ldots\ldots\ldots\ldots (1)$$

$$\frac{b}{\cos(200^\circ)} + \frac{d\sin(200^\circ)}{\cos(200^\circ)} = c \quad \ldots\ldots\ldots\ldots\ldots (2)$$

$a = d\cos(200^\circ) + c\sin(200^\circ) \quad \ldots\ldots\ldots\ldots\ldots (3)$

$b = c\cos(200^\circ) - d\sin(200^\circ) \quad \ldots\ldots\ldots\ldots\ldots (4)$

$\therefore\ a^2 + b^2 = c^2 + d^2$

$\therefore\ (a^2 + b^2 + c^2 + d^2) = 2\ (c^2 + d^2)$

$ac = dc\cos(200°) + c^2\sin(200°)$

$bd = cd\cos(200°) - d^2\sin(200°)$

Also _____

$ac - bd = (c^2 + d^2)\sin(200°)$

$$\text{So}\frac{a^2+b^2+c^2+d^2}{ac-bd} = \frac{2(c^2+d^2)}{(c^2+d^2)\sin(200°)} = 2\cos ec(200°)$$

$\therefore\ p = 2$

57.

$$I_n = \left(\frac{e^{-x}(\sin x)^n}{-1}\right)\Big|_0^\infty + n\int_0^\infty [(\sin x)^{n-1}\cos x e^{-x}dx$$

$$= 0 + n\left(\frac{(\sin x)^{n-1}\cos x\, e^{-x}}{-1}\right)\Big|_0^\infty$$

$$+ n\int_0^\infty [(\sin x)^{n-1}(-\sin x) + (\cos x)(n-1)\cos x(\sin x)^{n-2}\cos x]e^{-x}dx$$

$$= n\int_0^\infty (-(\sin x)^n + (n-1)(1-\sin^2 x)(\sin x)^{n-2})e^{-x}dx$$

$$= \frac{n(n-1)}{n^2+1}I_{n-2}$$

$$101\frac{I_{10}}{I_8} = 101\times\frac{10\times 9}{10^2+1} = 90.$$